The Legend Challenge: Embedding Ethical Compliance into LLMs through Champion Narratives in the Gender Equality Domain

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Abstract

Sargsyan and Damiani (2025) proposed that fine-tuning large language models on *legends* (synthetic narrative exemplars of compliant behaviour) may embed regulatory ethics more effectively than fine-tuning on the underlying regulatory text, but provided neither an empirical verification nor a measurement instrument. We provide both. We compare legend-based fine-tuning with regulation-text-based fine-tuning on a single regulation (the EU Gender Equality Strategy 2020–2025) and a single backbone (`gpt-4o-2024-08-06`), holding the system prompt, the training format and the JSONL line count constant, so that corpus content is the only experimental variable. To make the comparison measurable we introduce **GenderEqGLUE**, a five-task benchmark: GE-CLS, GE-NLI, GE-QA, GE-WSC and GE-NEXT, adapted one-to-one from the GLUE and SuperGLUE templates. To examine *how* each model arrives at its predictions we conduct an interpretability analysis based on the log-probabilities exposed by the OpenAI API for the fine-tuned models, measuring via **structured occlusion** how much each model depends on the scenario and on the regulatory clause over a stratified sample of 60 GE-NLI items. The analysis shows that both fine-tuned variants sharply reduce their dependence on explicit context relative to the unadapted model, and that `tuned-legends` depends on the regulatory clause almost three times more than `tuned-regulation` ($p = \mathbf{0.0001}$), with the divergence concentrated on the decidable labels and absent on `neutral`. On the aggregate GenderEqGLUE Score, `tuned-regulation` leads with 0.938, `tuned-legends` follows with 0.926 and `base` comes last with 0.904; per-task wins split 3–3 between the two fine-tuning regimes (`base` wins none). `tuned-legends` is the strongest model on the task that is central to the hypothesis, compliant-action selection, GE-NEXT. The two regimes therefore teach **complementary competences**: legends teach compliance recognition and compliant-action selection; the regulatory text teaches ontology recognition and factual recall of short text spans. The deployment choice should be driven by the application's dominant risk profile.

1 Introduction

Regulatory complexity and AI as a compliance instrument

The contemporary regulatory landscape for sustainability is extremely complex; as Sargsyan and Damiani (2025) observe, it is characterised by overlapping requirements, policy gaps and a consequent rapid expansion of new rules. The Corporate Sustainability Reporting Directive requires large firms to report on social matters, including gender equality policies (European Parliament and Council, 2022b); the EU Taxonomy Regulation obliges financial institutions to declare alignment

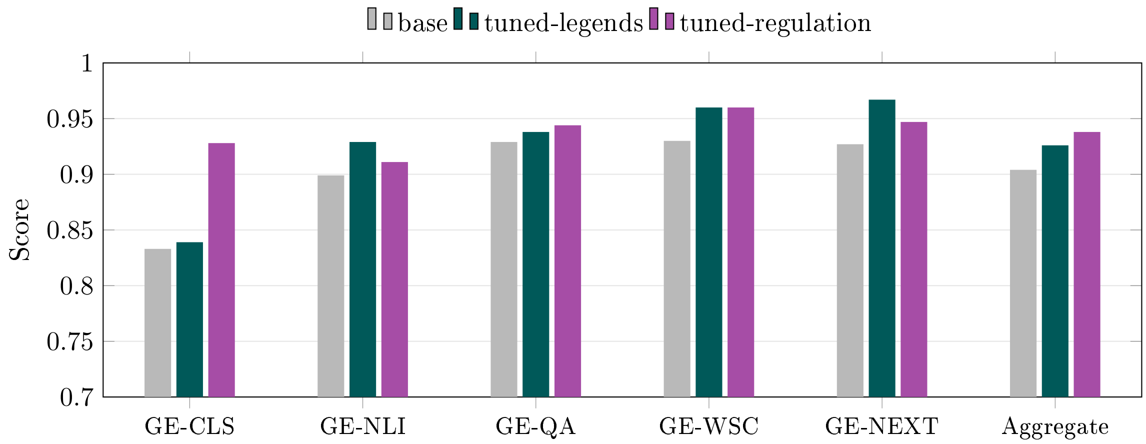


Figure 1: The two fine-tuning regimes teach complementary competences. `tuned-regulation` leads on the aggregate score; `tuned-legends` is the strongest model on GE-NEXT and GE-NLI.

percentages (European Union, 2020); the Women on Boards Directive imposes representation targets for the underrepresented gender on the boards of listed companies (European Parliament and Council, 2022a); and the Pay Transparency Directive establishes wage-transparency reporting obligations (European Parliament and Council, 2023). Superimposed on these public obligations are the internal policies that individual firms maintain on ethics, diversity and inclusion, all of which must be weighed in any single corporate decision (Sargsyan & Damiani, 2025). Generative AI is widely proposed as an instrument for translating sustainability requirements into operational decisions (Storey et al., 2025), yet the optimisation objective these models inherit is structurally narrow: cost reduction and profit maximisation.

The compliance paradox

Sargsyan and Damiani (2025) formalise this tension as the *compliance paradox*: when practical implementation is driven exclusively by profit maximisation, a conflict with regulatory intent arises. Their canonical illustration is a hiring scenario in which a firm committed to gender equality adopts a Human Resource Management module that, although designed to be "gender-blind", favours a male candidate whose profile satisfies only 80% of the stated requirements. The optimisation-driven decision contradicts both the firm's declared commitment and the underlying regulatory expectation. Sargsyan and Damiani (2025) further locate the implementation pressure point of this paradox at the level of middle management: "Critically, the direct implementation of policies and regulations by middle management and their potential focus on costs and profits creates a conflict of interest when AI implementation is delegated to them". The structural diagnosis is that middle managers, charged with operationalising both regulatory directives and corporate strategy, may rationally prioritise the cost objective—the one on which their performance is measured—over regulatory directives, whose enforcement is remote and probabilistic. The fact that by 2026 twenty per cent of organisations will use AI to halve middle-management roles (The Economist, 2024) only intensifies this tension, because the layer most exposed to regulation-versus-cost arbitrage is also the one most exposed to replacement by AI.

Prior approaches and their limitations

Two responses to this paradox appear in the literature. The first is *ex post*: firms make decisions independently and regulatory compliance is assessed after the fact. Sargsyan and Damiani (2025) argue that reactive delay is unacceptable for high-impact social outcomes. The second is *ex ante*: regulatory considerations are embedded directly into the AI system, for example through contextual regulatory alignment (Achintalwar et al., 2024) or machine-readable semantic formats (McLaughlin et al., 2021). These approaches require translating the regulation into a formal representation, a process whose difficulty across jurisdictions, languages and sources of authority is repeatedly reported in the digital-ready legislation literature (Motzfeldt et al., 2018).

The legends proposal and its empirical gap

Sargsyan and Damiani (2025) propose a third route: instead of formalising the regulation, they exploit it as a generator of *legends*—synthetic, idealised exemplars (*champion profiles*) of individuals or institutions whose behaviour embodies compliance. The training intuition, grounded in work on organisational testimonials (Kanyamukenge et al., 2024) and in findings that reliable or ideal training examples improve robustness (Northcutt et al., 2017), is that an LLM fine-tuned on legends will internalise an ethical inductive bias. The model is trained to recognise the patterns of compliant behaviour as reference exemplars, so that it can flag deviations even when the formal rule is not explicitly retrieved, acting as a judge whose assessment may be overridden by a human but is never structurally absent. The proposal is normatively attractive and mechanically specific, but it leaves a fundamental empirical question unanswered: *does fine-tuning on legends in fact produce a model measurably different from one obtained by fine-tuning on the regulatory text, all other conditions being equal?* No empirical evaluation, no measurement instrument and no controlled comparison accompany the original proposal.

A second gap is the absence of a domain-specific benchmark. Standard NLU suites such as GLUE (Wang et al., 2018) and SuperGLUE (Wang et al., 2019) are deliberately domain-independent; their tasks reward general linguistic competence and do not test whether a model recognises compliance, selects compliant actions or generalises over a theme such as gender equality.

Why gender equality?

Another point worth addressing in this introduction is that the proposal is general: it applies to any sustainability domain whose regulatory landscape is dense and whose social costs are hard to encode in a closed-form objective. Two areas nevertheless receive particular weight in their introduction: sustainable finance and gender equality. We select the gender-equality domain for three reasons. First, it has high semantic complexity: the EU Gender Equality Strategy 2020–2025 (European Commission, 2020) articulates numerical targets, institutional mechanisms and normative principles, requiring a model to reason simultaneously over quantitative, institutional and evaluative registers. Second, hiring, promotion, parental leave and pay equity are routinely mediated by human resource management software, making the cost-versus-fairness trade-off concrete and recurrent. Third, the EU *acquis* in this area comprises the Pay Transparency Directive (EU) 2023/970, the Work-Life Balance Directive (EU) 2019/1158, Directive (EU) 2022/2381 on women on boards, the Victims' Rights Directive 2012/29/EU and the Council of Europe's Istanbul Convention—a dense network that offers a broad held-out evaluation surface, thematically adjacent to the training regulation but distinct from it.

Contributions

We make three contributions which, taken together, support a preliminary verification of the legends hypothesis under matched fine-tuning conditions. *First*, we develop a **dataset-construction pipeline**, derived from SustainableQA (Wattenberg et al., 2025), that converts unstructured regulatory text and synthetic legends into matched fine-tuning corpora; the resulting JSONL files share system prompt, line count and chat format, leaving corpus content as the only experimental variable. *Second*, we introduce **GenderEqGLUE**, a five-task benchmark for gender-equality compliance reasoning: GE-CLS (pillar classification), GE-NLI (compliance entailment), GE-QA (regulation reading comprehension), GE-WSC (stereotype-aware coreference with a Gender Parity Score) and GE-NEXT (compliant-action selection with a four-category distractor typology). *Third*, we conduct an **interpretability analysis via structured occlusion** which, exploiting the log-probabilities exposed by the OpenAI API for the fine-tuned models, measures over 60 GE-NLI items how much each model depends on the scenario and on the regulatory clause; the analysis documents that fine-tuning reduces dependence on explicit context and that the two regimes internalise distinct objects — the content of the norm in one case, the application procedure in the other. Together, these three artefacts support the following claim, which we adopt as the minimal defensible reading of our results:

> When evaluated on a single backbone (`gpt-4o-2024-08-06`) fine-tuned through FineTuneDB Studio with size-matched JSONL corpora, the results on the GenderEqGLUE benchmark reveal that the two fine-tuning regimes teach complementary competences. The regulation-based regime obtained a 1.2-point advantage in aggregate score, winning mainly on the tasks that match the internal thematic and lexical structure of the regulation (GE-CLS, GE-QA on short text spans). However, the legends-based regime

demonstrated superior reasoning in applied scenarios: it improved on both the base model and the regulation model on GE-NEXT (compliant-action selection) and tied or outperformed on the two other tasks (GE-NLI and GE-WSC parity).

We deliberately restrict the scope to a single backbone (`gpt-4o-2024-08-06`), size-matched JSONL corpora and a single regulatory domain; behaviour outside this scope is discussed in Section 7. The remainder of the paper is structured as follows. Section 2 describes the dataset-construction pipeline. Section 3 details the construction of GenderEqGLUE. Section 4 reports the headline results and per-task breakdowns. Section 5 presents the interpretability analysis via structured occlusion and its results. Section 6 discusses the complementarity finding and its implications for deployment, and Section 8 concludes.

2 Methodology: Dataset-Construction Pipeline

2.1 Pipeline Overview

Empirically verifying the Sargsyan-Damiani hypothesis requires two fine-tuning corpora that differ only in the property of interest, namely whether their content is regulatory text or narrative exemplars derived from that text. The pipeline described in this section is the procedural instrument that produces these matched corpora; it is the methodological prerequisite for the controlled comparison reported in Section 4.

The remainder of this section presents each stage. Figure 2 summarises the end-to-end pipeline. The pipeline adapts and simplifies the **SustainableQA** (Wattenberg et al., 2025) framework to the scope of gender-equality compliance: this is a scalable pipeline for generating Q&A pairs from corporate sustainability reports, combining semantic chunk classification, hybrid span extraction and the generation of factoid and non-factoid questions for each span. Our final pipeline comprises seven stages: (1) preprocessing (PDF $\rightarrow$ Markdown conversion plus regex-based cleaning, Section 2.2); (2) legend generation (Section 2.3); (3) six-class pillar classification (Section 2.4); (4) two-stage span extraction (Section 2.5); (5) factoid and non-factoid Q&A generation (Section 2.6); and (6) fine-tuning of the GPT-4o backbone on the FineTuneDB Studio platform (Section 2.7). Legends and regulations pass through the same stages, share a single system prompt and emerge as parallel JSONL files with approximately matched line counts (327 versus 293) on which the comparison is run.

2.2 Preprocessing: from PDF to Clean Markdown

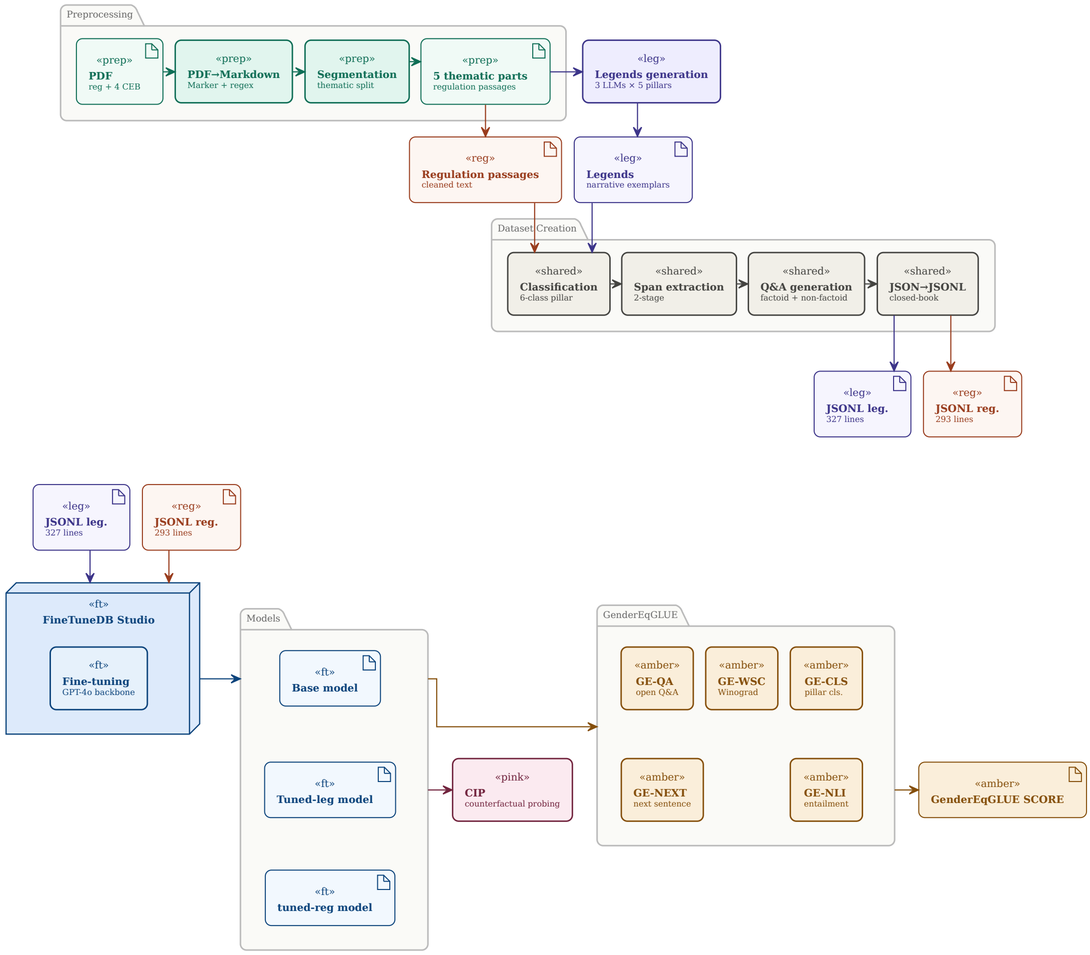
The regulation is published as a densely laid-out PDF containing footnotes, page-level URLs, image references and rendering artefacts. Downstream Q&A generation by the LLM requires semantically coherent passages of bounded length; any layout residue inflates the token budget and risks contaminating the generated questions with non-substantive material.

Preprocessing proceeds in two stages. First, the PDF is converted to Markdown using the Marker library (Paruchuri, 2024), which preserves the heading hierarchy of the source document. Second, a regex-based cleaning script removes inline footnote markers, full-text footnote blocks at page margins, image references, orphan URLs, language markers and consecutive blank lines. The cleaning stage reduces the document from 64,703 to 45,697 characters (**29.4% reduction**) while preserving all 16 section headings and every numerical commitment present in the source.

The cleaned regulation is then segmented in two stages. Stage 1 splits the document into five thematic parts using the top-level Markdown headings (`##`) as boundaries; we use the five resulting parts to extract five pillars with which we characterise the Strategy (violence_stereotypes, equal_economy, leadership_participation, mainstreaming_intersectionality, funding_global_action). Stage 2 further splits each thematic part along subheadings (`###`) and a maximum word-count constraint (max_words = 350). Splits occur at paragraph boundaries, so that no passage crosses a mid-sentence semantic break. Two-stage segmentation yields passages that are at once coherent enough to be processed with a single LLM call and short enough to remain within the context window of the downstream prompt; the table-handling stage of the SustainableQA framework is omitted, since neither the regulation nor the legends corpus contains tabular content.

2.3 Legend Generation

The Sargsyan-Damiani hypothesis requires *legends*, that is, narrative exemplars: short stories in which named characters demonstrate concrete compliance with a specific regulatory clause. These



Shape	Meaning
Rectangle	**Process** / transformation step
Artifact	**Data file** / output / input
Node	**Platform** / runtime
	Shared preprocessing
	Legends branch
	Regulation branch
	Shared stages
	Fine-tuning / models
	GenderEqGLUE
	Explainability (CIP)

Figure 2: End-to-end pipeline. The legends branch and the regulation branch pass through identical stages of preprocessing, classification, span extraction and Q&A generation, differing only in source content. The resulting JSONL files share system prompt, chat format and approximate line count, isolating corpus content as the only experimental variable.

exemplars are not present in the source regulation, which describes obligations in the abstract; they must be synthesised.

Three commercial LLMs, accessed through their respective web interfaces in April 2026—*GPT-5.4 Pro* (ChatGPT), *Gemini 3 Pro* and *Microsoft Copilot*[1]—each generate five legends, one per pillar, conditioned on the corresponding thematic part as context. Each model is governed by an identical system prompt that fixes the legend format (title, setting, 3–5 named characters, a 400–600 word narrative arc with at least one direct-speech dialogue, a measurable outcome and an acknowledgement of the EU Gender Equality Strategy 2020–2025); the user prompt for each call supplies the regulatory part as ⟨context⟩. The system prompt is reproduced in full in Listing 3. The output is 15 legends (5 pillars × 3 models).

[1] Model versions corresponding to the default settings of 10–15 April 2026

Multi-LLM mixing maximises stylistic and lexical diversity in the training set, reducing the risk that the legend fine-tuned model overfits the idioms of a single generator. The legend pool is small (15 narratives) but yields 327 closed-book Q&A pairs after the downstream extraction and generation stages (Section 2.6), a line count comparable to the 293 pairs derived from the regulatory text.

2.4 Six-Class Pillar Classification

Each passage coming out of the segmentation stage is classified by *Gemini 3 Pro* into one of six categories: the five pillars listed above plus an `unknown` rejection class.

The `unknown` class captures administrative metadata, document headings, reference lists and any passage that does not clearly belong to a pillar. The choice to include an explicit rejection class, rather than forcing every passage into one of the five pillars, is deliberate: without it, off-topic passages would be silently mislabelled and would contaminate downstream Q&A generation.

A single classifier and a single taxonomy are reused for (i) the regulatory-text JSONL, (ii) the legends JSONL and (iii) the CEB benchmark pool. This unification is the structural reason why performance differences across the five GenderEqGLUE tasks can be compared across pillars: the pillar labels mean the same thing in training and in evaluation.

2.5 Two-Stage Span Extraction

Factoid Q&A generation requires identifying short verbatim text spans that can serve as reference answers. SustainableQA's solution combines a pre-trained NER model with regex rules and a post-hoc verification stage. This stack is justified for noisy corporate documents but is unnecessarily heavy for clean regulatory prose and synthetic narratives.

We replace it with a single two-stage LLM-driven extractor. Stage 1 is a high-recall extractor (Listing 4): Gemini 3 Pro is asked to identify every verbatim text span in a passage that aligns with the pillars of the regulation, under strict length constraints (1–5 words typically, up to 8 for complex named entities). Stage 2 performs three operations on the candidate spans (Listing 5): contextual verification (ensuring that each span appears verbatim in the source), filtering (removing redundancies) and thematic clustering (grouping semantically related spans under a descriptive label).

The two-stage decomposition separates concerns. Stage 1 maximises recall without worrying about uniqueness; Stage 2 enforces verbatim presence and groups the verified spans into clusters that can serve as multi-span answers to descriptive questions. The output of thematic clustering also feeds the non-factoid Q&A generator in Section 2.6, which benefits from groups of related spans when constructing relational questions.

2.6 Q&A Generation: Factoid and Non-Factoid

The fine-tuning corpus must cover two question modes: (i) factual recall of short spans (factoid), which probes the model's memorisation of concrete commitments; and (ii) descriptive reasoning (non-factoid), which probes its understanding of relationships, mechanisms and implications. A single question type would teach a single capability; both are required to support the five downstream benchmark tasks.

For each passage two prompts are issued to *Gemini 3 Pro* through the Gemini web interface. The factoid generator (Listing 6) takes the verified span output of Section 2.5 and produces questions whose answer is exactly one verbatim text span. The non-factoid generator (Listing 7) produces descriptive answers (1–4 sentences) grounded in the passage, with a mandatory four-rule self-correction checklist (genuinely non-factoid, distinct from the others, fully answerable from the context, accurate summary of the context). Both generators output JSON. Representative examples are shown in Appendix B.

2.7 Fine-Tuning Protocol

Each JSONL line contains three chat turns: a system prompt anchoring the model to the gender-equality domain (identical for both corpora); a user turn containing the question; an assistant turn containing the reference answer. The system prompt is:

Listing 1: Fine-tuning system prompt.

```
You are an expert assistant on the EU Gender Equality
Strategy 2020--2025. Answer questions about the strategy's policy objectives,
instruments, and concrete examples of compliance accurately and concisely,
grounded in the Strategy's text and its illustrative legends.
```

A single training line therefore has the structure shown in Appendix B.

The controlled comparison requires that the two fine-tuning runs differ *only* in the content of their respective JSONL files.

Both runs use the same backbone, `gpt-4o-2024-08-06`, fine-tuned through the FineTuneDB Studio interface. The legends JSONL contains 327 lines (199 non-factoid + 128 factoid, with 45 `unknown` chunks discarded) distributed across the five pillars as in Table 1. The line counts are within ≈10% of each other, the closest match achievable given the distinct content distributions of the two source corpora.

Table 1: Dataset statistics and per-pillar distribution for the Legends and Regulation runs (Backbone: `gpt-4o-2024-08-06`)

Pillar	Legends	Regulation
`equal_economy`	64	90
`funding_global_action`	68	50
`leadership_participation`	62	40
`mainstreaming_intersectionality`	66	27
`violence_stereotypes`	67	86
`unknown` chunks	45	5
Non-factoid	199	184
Factoid	128	109
Total lines	327	293

By using FineTuneDB Studio for both runs, hyperparameter selection, optimiser choice, learning-rate schedule and termination criteria are externally controlled and identical across the two runs.

3 GenderEqGLUE: A Regulatory Compliance Reasoning Benchmark

3.1 Overview

The standard NLU benchmark suites, GLUE (Wang et al., 2018) and SuperGLUE (Wang et al., 2019), provide composite scores over single-sentence classification, NLI, reading comprehension and commonsense reasoning, with constituent datasets including SST-2, MNLI, RTE, SQuAD (Rajpurkar, Zhang, Lopyrev, & Liang, 2016), BoolQ (Clark et al., 2019), WSC, COPA (Roemmele, Bejan, & Gordon, 2011) and the gender-aware coreference probe WinoBias (Zhao, Wang, Yatskar, Ordonez, & Chang, 2018). Their design choice is deliberately domain-independent; a high GLUE score reflects generic linguistic competence, not reasoning about the structure of a specific regulation or domain compliance reasoning. Testing the Sargsyan-Damiani hypothesis requires a benchmark that probes the competence the hypothesis specifically speaks about. We build one. GenderEqGLUE adapts, one to one, the five canonical GLUE/SuperGLUE task templates to the gender-equality domain: single-sentence classification, natural language inference, reading comprehension, coreference and commonsense action selection. Table 2 summarises the five tasks.

The aggregate **GenderEqGLUE Score** is the unweighted arithmetic mean of the five headline per-task metrics, mirroring the original GLUE score.

3.2 Common Evaluation Base

Three of the five tasks (GE-CLS, GE-NLI, GE-QA) draw their items from a single **Common Evaluation Base**. The CEB is built from four EU documents thematically adjacent to the Strategy but absent from the training corpora: the Roadmap for Women's Rights (2025), the Gender Action Plan III (2021–2025), the Council Conclusions on closing the gender pay gap (June 2019) and Directive (EU) 2022/2381 on gender balance among directors of listed companies. These documents

Table 2: Overview of the GenderEqGLUE tasks. The first three tasks share a Common Evaluation Base (CEB) of 217 EU passages held out from fine-tuning; GE-WSC uses WinoBias (Zhao et al., 2018) as-is; GE-NEXT is built from scratch.

Task	Description	GLUE analogue	Source	Metric
GE-CLS	Pillar classification (6 classes)	SST-2	CEB-CLS (72 passages)	Macro-F1
GE-NLI	Compliance entailment (3 classes)	MNLI / RTE	CEB-NLI	Accuracy
GE-QA	Regulation reading comprehension	SQuAD / BoolQ	CEB-QA (open-book)	F1 / EM, accuracy
GE-WSC	Stereotype-aware coreference	WSC / Winogender	WinoBias	Accuracy + parity
GE-NEXT	Compliant-action selection (4 choices)	COPA	Curated vignettes	Accuracy

pass through the same preprocessing pipeline as the Strategy, are classified by the same six-class pillar classifier and are stratified into three disjoint subsets (CEB-CLS, CEB-QA, CEB-NLI). Table 3 reports the per-pillar distribution.

Table 3: Common Evaluation Base: per-pillar distribution after the 1/3 stratified split. The `unknown` class is excluded from the construction of the substantive tasks but retained as a rejection target for GE-CLS.

Pillar	Pool	CLS	QA	NLI
`violence_stereotypes`	72	24	24	24
`leadership_participation`	45	15	15	15
`funding_global_action`	21	7	7	7
`equal_economy`	18	6	6	6
`mainstreaming_intersectionality`	11	4	3	4
`unknown`	50	16	17	17
Total	217	72	72	73

3.3 Task 1, GE-CLS (Pillar Classification)

GE-CLS asks the model to classify a passage into one of the five gender-equality pillars or as `unknown`, framing the problem as six-way single-input classification with Macro-F1 as the headline metric, supplemented by per-class F1 and a confusion matrix as diagnostics. The data come directly from `CEB-CLS` (72 passages), since each passage already carries its pillar label from the CEB classification stage and no additional construction is required. In design terms, GE-CLS preserves the surface mechanics of SST-2, mapping a single textual input to a categorical label without auxiliary context, but generalises the decision from binary sentiment polarity to six-way thematic classification: where SST-2 evaluates the internalisation of affective valence, GE-CLS evaluates the internalisation of *topical structure*, that is, whether the model has acquired the pillar ontology that the regulation imposes on the domain.

3.4 Task 2, GE-NLI (Compliance Entailment)

Given a scenario (premise) describing organisational behaviour and a regulatory clause (hypothesis), GE-NLI asks the model to decide whether the scenario *entails*, *contradicts* or is *neutral with respect to* the clause, framing the problem as three-class classification with accuracy as the headline metric; this is one of the **central tasks** for the Sargsyan-Damiani hypothesis, since it directly measures *compliance recognition*, the competence legends are designed to teach. The dataset consists of 168 triples (56 per class), generated from the 56 `CEB-NLI` passages remaining after filtering the 17 `unknown` passages, through a five-stage construction process: for each passage, Stage 1 extracts a regulatory clause as the hypothesis (length 8–35 words, falsifiable by an organisational scenario); Stage 2 generates a compliant fictional scenario as the entailment premise; Stage 3 perturbs each compliant premise along a single quantitative or factual axis (numeric inversion, policy reversal, threshold undershoot, mechanism removal, scope reduction, deadline miss), keeping the hypothesis unchanged, to produce a contradiction triple; Stage 4 pairs each compliant premise (pillar A) with a hypothesis from a different pillar (pillar $B \neq A$) to produce a neutral triple; and Stage 5 verifies class balance (56/56/56) and runs deduplication. The resulting dataset contains one line per triple in the schema shown in Appendix B.

3.5 Task 3, GE-QA (Regulation Reading Comprehension)

GE-QA asks the model to produce the correct answer given a regulatory passage and a question, and is split into two sub-tasks: GE-QA-Factoid (SQuAD-style) produces extractive span answers scored by F1 and Exact Match, while GE-QA-Bool (BoolQ-style) produces yes/no answers scored by accuracy, and the two sub-task metrics are averaged into a single GE-QA score. GE-QA is the only *open-book* task in the benchmark; the passage is supplied in the user prompt at evaluation time, and this is a deliberate design choice, since the closed-book tasks (GE-CLS, GE-NLI, GE-NEXT) test whether fine-tuning has internalised the regulation, whereas GE-QA tests whether the model can locate answers when the regulation is supplied, making the open/closed-book contrast across the benchmark itself a diagnostic instrument for distinguishing internalisation from reading-skill effects. On the data side, GE-QA-Factoid reuses the two-stage span extractor and the factoid generator of Sections 2.5 and 2.6 on `CEB-QA`, with one adaptation: questions must be *self-contained* (no "the passage", "this text" or meta-linguistic cues), and spans longer than 7 words are discarded; GE-QA-Bool is constructed with two methods balanced 50/50, *direct paraphrase* (a passage claim rewritten as a polar question, gold = yes) and *perturbed claim* (a single quantitative or scope dimension inverted, gold = no).

3.6 Task 4, GE-WSC (Stereotype-Aware Coreference)

GE-WSC asks the model to resolve pronominal coreference in professional contexts without relying on gender stereotypes, framing the problem as binary classification and reporting two metrics: (i) accuracy averaged across the four WinoBias subsets (type1_pro, type1_anti, type2_pro, type2_anti), and (ii) a **Gender Parity Score** = $|\,\mathrm{acc(pro\text{-}stereotype)} - \mathrm{acc(anti\text{-}stereotype)}\,|$, where a value of 0 indicates stereotype-invariant behaviour. WinoBias (Zhao et al., 2018) is used *as-is* from the HuggingFace Hub (`uclanlp/wino_bias`), with all four subsets preserved unmodified and no construction performed on our side, a choice that keeps GE-WSC directly comparable with the broader NLP literature on bias evaluation.

3.7 Task 5, GE-NEXT (Compliant-Action Selection)

GE-NEXT presents the model with a vignette describing a quantified gender-equality gap and asks it to select the next-step action most consistent with the gender-equality domain, distinguishing *substantive compliance* from three distractor types, in a four-option multiple-choice format in which each item consists of a 2–4 sentence vignette identifying a fictional middle-management protagonist, an organisational sector and size, a concrete gap quantified in numerical or structural terms, and four candidate actions `A`–`D`, exactly one of which is the **substantive-compliant** gold, while the three distractors are drawn from a fixed typology comprising **performative** options (a symbolic gesture without measurable KPIs, such as public statements or generic awareness campaigns), **cost-optimising** options (a response that defers, minimises or absorbs the cost of remediation at the expense of the regulatory objective, such as postponing remediation or capping the budget at a fraction of the identified gap) and **orthogonal** options (a plausible organisational action addressing a *different* gender-equality concern than the one in the vignette, such as commissioning diversity training when the gap is a pay-transparency violation); accuracy is the headline metric, with per-pillar accuracy and per-distractor-type error rate reported as diagnostics. The dataset is built from scratch (150 items, 30 per pillar) because no public dataset provides the action-selection format against a regulatory anchor: Stage 1 generates vignettes conditioned on (pillar label, regulatory clause from the CEB, fictional middle-manager constraint), Stage 2 generates the four options per vignette, and Stage 3 shuffles option positions with a fixed seed to uniformise the gold-letter distribution across A–D; an example item is shown in Appendix B. GE-NEXT is the most direct probe of the compliance-versus-cost arbitrage that Sargsyan and Damiani (2025) place at the centre of the implementation conflict, since every item forces a choice between substantive compliance and three structured alternatives, whereas the regulatory text, by contrast, describes obligations but never models the *rule-versus-cost trade-off at the point of decision*, so that GE-NEXT tests whether legend fine-tuning has embedded not just the *content* of the regulation but also the *decision pattern* that compliance requires.

3.8 Aggregate Score and Statistical Protocol

For each model, `base`, `tuned-legends`, `tuned-regulation`, we report the five headline per-task metrics and their unweighted arithmetic mean (the **GenderEqGLUE Score**).

4 Results

4.1 Headline Scores and Per-Task Wins

Table 4 reports the five headline per-task metrics and the aggregate GenderEqGLUE Score.

Table 4: Headline per-task metrics and aggregate GenderEqGLUE Score. **Bold red**: column maximum. Ties on GE-WSC are bolded for both fine-tuned regimes.

Model	GE-CLS (Macro-F1)	GE-NLI (Accuracy)	GE-QA ((F1+Bool)/2)	GE-WSC (Accuracy)	GE-NEXT (Accuracy)	**GenderEqGLUE** (mean)
`base`	0.833	0.899	0.929	0.930	0.927	0.904
`tuned-legends`	0.839	**0.929**	0.938	**0.960**	**0.967**	0.926
`tuned-regulation`	**0.928**	0.911	**0.944**	**0.960**	0.947	**0.938**

`tuned-regulation` **is the numerically strongest model in aggregate**, with a GenderEqGLUE Score of 0.938, a margin of +3.4 points over the base model and +1.2 points over `tuned-legends`. Both fine-tuned models improve markedly over the base model. The aggregate ranking is `tuned-regulation` $\approx$ `tuned-legends` > `base`.

The split of per-task wins is informative: `tuned-regulation` wins on the tasks that directly probe the thematic ontology of the regulation and the factual recall of short spans (GE-CLS, GE-QA); `tuned-legends` wins on the two tasks designated *a priori* as the central tests of the Sargsyan-Damiani hypothesis (GE-NLI on compliance recognition, GE-NEXT on compliant-action selection). The aggregate score *hides* this split, which is why we read it alongside the per-task results rather than in their place.

4.2 Pillar Effects Across Tasks

Table 5 reports the per-pillar performance *gains* over the base model for the four tasks where pillar labels are available (GE-WSC is excluded since it uses WinoBias professions, not EU pillars).

4.3 GE-WSC: Gender Parity Diagnostics

The two fine-tuned models tie on aggregate GE-WSC accuracy (0.96 versus 0.96, against 0.93 for the base model), but the Gender Parity Score reveals a qualitative difference. `tuned-legends` achieves a perfect parity score of 0.000 (identical accuracy on the _pro and _anti subsets for both WinoBias types); `tuned-regulation` achieves a mean parity of 0.040, with a parity of 0.080 on Type-1 (slightly *worse* than the base model's 0.040) and 0.000 on Type-2. The accuracy gain of `tuned-regulation` comes entirely from the _pro items, which widens the pro/anti gap on Type-1 — exactly the asymmetric improvement that the parity score is designed to detect. The two tuned models therefore reach the same accuracy through qualitatively different routes; only `tuned-legends` satisfies the fairness criterion that the parity diagnostic formalises.

Table 5: Pillar effects across tasks. Each cell is the per-pillar performance gain (winning fine-tuned model − base) on the indicated task. The larger gains on the minority pillars (`equal_economy`, `mainstreaming_intersectionality`) should be read bearing in mind their small sample sizes ($n = 6$, $n = 4$).

Pillar	GE-CLS (reg − base)	GE-NLI (legends − base)	GE-QA (reg − base)	GE-NEXT (legends − base)
`violence_stereotypes`	+0.045	+0.028	+0.023	+0.067
`equal_economy`	+0.167	+0.111	+0.029	+0.067
`leadership_participation`	+0.000	+0.000	+0.049	+0.000
`mainstreaming_intersectionality`	+0.389	+0.000	+0.025	+0.033
`funding_global_action`	+0.000	+0.048	+0.063	+0.033

5 Interpretability Analysis via Structured Occlusion

5.1 Motivation and Access Context

The benchmark results in Section 4 quantify *what* each model produces on the five tasks but are silent on *why*. The Sargsyan-Damiani hypothesis does not merely claim that legend-based fine-tuning improves metrics: it claims that legends embed the regulatory logic of the regulation into the model's decision structure as an internalised inductive bias, rather than as a surface lexical pattern. Empirically distinguishing *structural embedding* from *lexical mimicry* requires opening up the model's prediction and inspecting which portions of the input it actually uses.

An analysis of this kind rests on attribution methods that assign to each portion of the input a measure of its contribution to the prediction. Such methods fall into two families depending on the level of access they require. *White-box* methods need the model's internal components: **integrated gradients** integrate gradients from a baseline to the input, while **attention rollout** inspects attention weights across the transformer stack. *Black-box* methods, by contrast, require no internal access and operate by perturbing the input and observing how the output changes.

The applicability of each family depends on the degree of model access. The three models compared (`base`, `tuned-legends`, `tuned-regulation`) share the `gpt-4o-2024-08-06` backbone, a closed proprietary model, and can be queried programmatically on the OpenAI endpoint. That endpoint returns the *log-probabilities* of the generated tokens, but exposes neither gradients nor attention matrices. It follows that white-box methods and attention visualisation are not applicable, whereas black-box methods, which require only the ability to query the model and observe its output confidence, are fully applicable. It is on this family that the following analysis is built.

From token to structural field.

An initial exploratory campaign applied occlusion at the level of the individual token on samples from GE-CLS, GE-NLI and GE-NEXT. The outcome was negative for a structural reason: on passages of several hundred words, removing a single token does not appreciably change the confidence of a model that assigns the correct label a probability close to one. The signal was diluted across hundreds of positions and essentially indistinguishable from numerical noise, to the point that on GE-NEXT all measured importances were zero. This observation motivated a redesign of the definitive analysis along two axes. First, the unit of occlusion was raised from the token to the *structural field* of the item, that is, to a portion of the input with an explicit functional role. Second, the analysis was concentrated on GE-NLI alone, the only task whose two-field structure — premise and hypothesis — makes field-level occlusion directly interpretable in terms of the research hypothesis.

5.2 Methodology

The GE-NLI task presents each model with a pair consisting of an organisational **scenario** (the premise) and a **regulatory clause** (the hypothesis), and requires it to establish whether the latter is entailed, contradicted or left indeterminate by the former. The two fields have distinct epistemic statuses: the premise is factual material that the model encounters for the first time at inference, whereas the hypothesis is regulatory material that fine-tuning may, in principle, have internalised.

The measure of how much each model depends on the explicit presence of each field is therefore a direct measure of the degree of internalisation of the norm.

The signal is the *log-probability* that the model assigns to the correct label. For each item, each model and each occlusion unit, the unit is removed from the input and the item is resubmitted to the model; the importance of the unit is defined as the drop in the log-probability of the correct label relative to the full input. A high value indicates that removing that portion degrades the answer, and therefore that it is causally relevant to the prediction; a value close to zero indicates that the model reaches the same conclusion with the same confidence even in its absence. Occlusion is applied at two nested levels: at the level of **whole fields**, alternately removing the entire premise and the entire hypothesis, and at the level of **individual sentences** within each field. The system prompt used at query time is the generic fine-tuning one (Section 2), with the task instructions placed in the user message, so as to reproduce faithfully the configuration in which the models were evaluated in Section 4.

The comparison between models is conducted with the Wilcoxon signed-rank test for paired samples, appropriate because the three models are evaluated on the same items and because the importance distributions are markedly skewed and non-normal. Given the skew, the **median** is reported alongside the mean, as a statistic more robust to the floor effects discussed below.

Note.

The **input** is a stratified sample of 60 GE-NLI items, 20 for each of the three labels, drawn from the Common Evaluation Base (Section 3.2). The **process** involves, for each item and each of the three models, one query on the full input, two queries for field occlusion and one query per sentence, reading the log-probabilities returned by the API. The **output** consists, for each item and each model, of the field and sentence importances, aggregated by label and submitted to the Wilcoxon test.

5.3 Results

Table 6 reports the importance of the two fields aggregated over the whole sample. The picture that emerges is sharp and organised on two levels.

Table 6: Mean and median importance of the two GE-NLI fields, measured as the drop in the log-probability of the correct label following occlusion of the field ($n = 60$). The p column reports the paired Wilcoxon test between `tuned-legends` and `tuned-regulation`.

	`base`		`tuned-legends`		`tuned-regulation`		
Field	mean	median	mean	median	mean	median	p
Premise	6.55	7.90	1.58	1.48	1.39	1.24	0.349
Hypothesis	7.28	8.16	4.11	3.59	1.38	0.55	**0.0001**

Fine-tuning reduces dependence on explicit context.

The first effect is common to both fine-tuned variants and separates them sharply from the unadapted model. `base` depends substantially on the presence of both fields: removing the premise or the hypothesis produces log-probability drops on the order of seven to eight units, a sign that the model builds its answer from the material present in the prompt. In the two adapted models this dependence falls by a factor of between two and five, and the difference relative to `base` is highly significant on both fields ($p < 0.001$ in all comparisons). Fine-tuning, regardless of the material on which it is conducted, shifts part of the decision basis from the prompt to the model's weights: it is the quantitative signature of the internalisation that the *closed-book* protocol (Section 2.7) was designed to induce.

The two variants internalise different things.

The second effect separates `tuned-legends` from `tuned-regulation` and is concentrated entirely on the hypothesis. On the premise the two models are statistically indistinguishable (1.58 versus

1.39; $p = 0.349$): both need the scenario to the same extent, which is expected, since the scenario is new factual material that no fine-tuning can have memorised. On the hypothesis, by contrast, the divergence is large and highly significant: `tuned-legends` depends on the regulatory clause almost three times more than `tuned-regulation` (4.11 versus 1.38; $p = 0.0001$). `tuned-regulation`, trained on raw regulatory text, reaches the correct label even when the clause is withheld, having evidently absorbed its content into its weights; `tuned-legends`, trained on compliance narratives, has instead internalised the *procedure* of comparing scenario against norm, and to apply it requires that the norm be presented to it.

Table 7: Hypothesis importance disaggregated by label ($n = 20$ per label). The divergence between the two fine-tuning regimes concentrates on the decidable labels and vanishes on `neutral`.

	`base`		`tuned-legends`		`tuned-regulation`		
Label	mean	median	mean	median	mean	median	p
`entailment`	7.97	11.89	5.55	6.00	2.18	0.81	**0.012**
`contradiction`	10.98	12.00	5.67	6.30	0.90	0.28	**0.0002**
`neutral`	2.90	1.75	1.11	0.59	1.06	0.26	0.498

The divergence is restricted to the decidable labels.

The per-label breakdown in Table 7 circumscribes the phenomenon further. On the two labels that require a substantive compliance judgement — `entailment`, where the scenario satisfies the norm, and `contradiction`, where it violates it — the gap between `tuned-legends` and `tuned-regulation` is large and significant. On `neutral`, where the norm is neither satisfied nor violated and the correct answer consists in recognising the indeterminacy, the gap vanishes entirely (1.11 versus 1.06; $p = 0.498$) and all three models show low importances. The reading is coherent: a judgement of indeterminacy rests on the *absence* of a link between scenario and norm, and therefore does not require the close examination of the norm that the two decidable labels impose. The fact that the divergence appears exactly where the task is substantive, and disappears where it is not, is what makes the effect interpretable as a difference in decision strategy rather than as a measurement artefact.

5.4 Limitations of the Analysis

Three limitations circumscribe the scope of these results.

First, the **saturation** observed in the exploratory campaign has not disappeared: on 37 of the 60 items all three models assign the correct label a probability close to one on the full input. Field-level occlusion remains informative because it removes large portions of the input, sufficient to shift confidence even in a saturated regime, but the available dynamic range is nonetheless compressed, and this attenuates rather than amplifies the measured differences: the reported estimates are therefore conservative.

Second, some `base` means are affected by a **floor effect**. The log-probability returned by the API is truncated at a minimum value, and the `base` means on `entailment` and `contradiction` (7.97 and 10.98, with medians of 11.89 and 12.00 respectively) are partly determined by items that collapse onto that threshold. The absolute value of these means should therefore be read as a lower bound on the actual dependence; it is for this reason that medians are reported alongside means, and for this reason that the statistical comparisons are conducted with a non-parametric rank-based test, insensitive to tail compression.

Third, the **sentence-level** occlusion produced no usable signal: individual sentence importances fall almost entirely below 0.5 and do not order sentences stably across models. Removing one sentence from a three- or four-sentence scenario leaves the model enough material to reach the same conclusion, and the phenomenon is the same, attenuated, one that made token-level occlusion unusable. Field granularity is therefore the finest at which this protocol produces interpretable attributions on the present setup.

To these is added the access limitation already discussed: the absence of gradients and attention matrices precludes any white-box verification of the attributions obtained. A follow-up study with a less capable backbone — which would reduce saturation by widening the dynamic range — and with a larger item pool is the study that these observations motivate.

Synthesis.

Taken together, the results describe a picture consistent with the complementarity hypothesis that emerged from the benchmark. Fine-tuning, in both variants, reduces the model's dependence on explicit context, and the two regimes internalise different objects: `tuned-regulation` absorbs the *content* of the norm, to the point of being able to do without it in the prompt; `tuned-legends` absorbs the application *procedure* and continues to require the norm as a term of comparison. Neither dominates the other: they learn distinct and measurably separate competences.

6 Discussion

6.1 Where the Hypothesis Is Supported

The Sargsyan-Damiani prediction that fine-tuning on legends produces a model with stronger compliance reasoning than fine-tuning on the regulatory text is supported in its central formulation. On GE-NEXT, the most direct probe of compliant-action selection, `tuned-legends` is the strongest model on the task. On GE-NLI, the compliance-recognition task, the result is directionally consistent: `tuned-legends` leads with an accuracy gain of +3.0 points over the base model, although the test size ($n = 168$) leaves the gap below significance. On GE-WSC the two fine-tuned models tie at the maximum, but the Gender Parity Score diagnostic distinguishes them: `tuned-legends` achieves perfect parity of 0, whereas the gain of `tuned-regulation` is asymmetric (pro-stereotype accuracy on Type-1 rises while anti-stereotype accuracy does not).

The interpretability analysis via structured occlusion (Section 5) provides a further layer of support that aggregate scores cannot offer. `tuned-legends` depends on the regulatory clause almost three times more than `tuned-regulation` ($p = 0.0001$), and the gap concentrates exactly on the labels that require a substantive compliance judgement, vanishing on `neutral`. The model trained on legends has not memorised the text of the norm — it has learned the procedure by which the norm is to be compared with a scenario, and to execute it requires that the norm be presented to it. This operating mode is consistent with what the Sargsyan-Damiani framing predicts as the product of training on narrative exemplars: not an archive of clauses, but an application procedure.

6.2 Where the Hypothesis Is Not Supported

The hypothesis does not generalise to ontology classification or to factual recall of short text spans. GE-CLS, which probes whether the model has internalised the pillar taxonomy, is dominated by `tuned-regulation` (0.928 Macro-F1 versus 0.839 for `tuned-legends`): the regulatory text explicitly mirrors the pillar structure, whereas the legends do not. GE-QA-Factoid is similarly favourable to `tuned-regulation` on short-span items, where the tendency of the legend-trained model to over-narrate depresses Exact Match more than F1 (a format misalignment effect rather than a knowledge deficit, but a deficit on the metric nonetheless). The interpretability analysis likewise circumscribes its own scope: the divergence between the two regimes emerges on the hypothesis but not on the premise, and on the decidable labels but not on `neutral`; moreover, over much of the sample the three models operate in a regime of confidence saturation, which compresses the dynamic range within which the effect can be measured.

These are not so much failures of the hypothesis as limits of its scope. Sargsyan and Damiani's claim is that legends teach compliance reasoning, not that they teach *everything*. The benchmark documents both halves of the picture; the discussion treats the second as informative, not as falsifying.

6.3 Complementarity, Not Dominance

The clearest synthesis of the headline results and the interpretability analysis is that legends and regulation teach **complementary competences**. Table 8 assigns each reasoning competence to the model that performs it best.

The practical implication for deployment is direct. A deployment whose risk profile foregrounds compliance recognition, compliant-action selection or fairness invariance has a defensible reason to prefer `tuned-legends` despite the 1.2-point deficit in aggregate score. A deployment that prioritises

Table 8: Complementarity assignment: which model is best at which reasoning competence. The assignment is derived from the per-task results (Table 4), the parity diagnostic (Section 4.3) and the interpretability analysis via structured occlusion (Section 5.3).

Reasoning competence	Best model	Evidence
Recognising compliance (entailment)	`tuned-legends`	GE-NLI 0.929 vs 0.911
Selecting the compliant action	`tuned-legends`	GE-NEXT
Detecting violations (contradiction)	`tuned-regulation`	GE-NLI contradiction subset
Classifying into the regulation ontology	`tuned-regulation`	GE-CLS 0.928 vs 0.839
Long-answer factoid extraction	`tuned-legends`	GE-QA-Factoid long-answer subset
Short-span factoid extraction	`tuned-regulation`	GE-QA-Factoid short-span subset

ontology recognition or concise factoid extraction would prefer `tuned-regulation`. The choice is governed by which competence the application's failure modes place first, not by the aggregate.

7 Limitations

We record the limitations of this study in the spirit of a declared scope rather than a concealed caveat.

Test set sizes are small. GE-CLS ($n = 72$), GE-NLI ($n = 168$), GE-QA-Factoid ($n = 123$), GE-WSC ($n = 100$), GE-NEXT ($n = 150$). The interpretability analysis in turn rests on a stratified sample of 60 GE-NLI items: sufficient to support the paired tests reported in Section 5, but limited to a single benchmark task, so that the extension of its conclusions to the other four remains to be verified.

Single backbone. The hypothesis is tested on `gpt-4o-2024-08-06`. Whether the legends advantages on GE-NEXT and GE-NLI extend to other backbones or to larger fine-tuning corpora is an open question that the present design cannot resolve. Backbone generalisability should be verified subsequently.

Discriminative ceiling. The backbone is already highly capable on some tasks: GE-WSC near the ceiling (0.93 base) leaves little headroom; GE-NEXT for `tuned-legends` is likewise near the ceiling at 0.967 (5 errors out of 150). Harder coreference probes (GAP, BUG) and harder GE-NEXT items would separate the models more sharply. The confidence saturation observed in the interpretability analysis — with all three models certain of the correct label on 37 of 60 items — is itself evidence that the discriminative ceiling against which fine-tuning effects can be measured is compressed.

Latent affinity between GE-NEXT synthetic text and the LLM evaluator. The vignettes and options are LLM-generated. Mitigation measures were applied (multi-LLM generation, manual paraphrasing of a QC sample, position shuffling with a fixed seed), but a residual latent affinity between generator and evaluator models cannot be excluded. Cross-model comparisons remain valid since all three models face identical items; *absolute* accuracy on GE-NEXT should be read as indicative rather than definitive.

The data and access deficit. The most significant operational limitation of this study is the combination of small test sets, a single backbone and closed platform access. A larger annotated benchmark, multi-backbone replication and access to the model's internals would, separately and above all together, refine every result reported here.

8 Conclusion

We have presented the first empirical verification of the Sargsyan-Damiani hypothesis that fine-tuning a large language model on narrative exemplars ("legends") derived from a regulation embeds regulatory compliance more effectively than fine-tuning on the regulatory text itself. The verification rests on three artefacts: a dataset-construction pipeline derived from SustainableQA that produces matched fine-tuning corpora; **GenderEqGLUE**, a five-task benchmark anchored in the gender-equality domain and adapted one to one from the GLUE and SuperGLUE templates; and an **interpretability analysis via structured occlusion** which, exploiting the log-probabilities exposed by the API of the fine-tuned models, measures each model's dependence on the scenario and on the regulatory clause, and thereby localises the divergence between the two fine-tuning regimes.

Three conclusions follow from the data. First, `tuned-regulation` leads on the aggregate GenderEqGLUE Score (0.938 versus 0.926 for `tuned-legends`, versus 0.904 for `base`) and `tuned-legends` is competitive; both clearly beat the base model. Per-task wins split 3–3 between the fine-tuned regimes. Second, the Sargsyan-Damiani hypothesis is supported in its central formulation: on GE-NEXT, the most direct probe of compliant-action selection, `tuned-legends` outperforms `base`, and the directional GE-NLI result, the GE-WSC parity diagnostic, the GE-QA long-answer subset and the occlusion-based interpretability analysis all corroborate it. Third, the hypothesis does not generalise to ontology classification or to factual recall of short spans: legends and regulation teach **complementary competences**.

A larger GenderEqGLUE v2, with ≥ 500 items per task, harder GE-NEXT items, harder coreference probes and multi-backbone replication, would convert the directional GE-NLI and GE-NEXT results into more solid evidence, resolve the legends-versus-regulation gap on GE-NEXT and test backbone generalisability. The framework presented here is designed to scale towards that effort.

References

Achintalwar, N., et al. (2024). [titolo su contextual regulation alignment]. *[Rivista]*, ,

Clark, C., Lee, K., Chang, M.-W., Kwiatkowski, T., Collins, M., Toutanova, K. (2019). BoolQ: Exploring the surprising difficulty of natural yes/no questions. *Proceedings of the 2019 conference of the north american chapter of the association for computational linguistics (naacl)* (pp. 2924–2936).

European Commission (2020). *A union of equality: Gender equality strategy 2020–2025.* COM(2020) 152 final. (https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52020DC0152)

European Parliament and Council (2022a). *Directive (eu) 2022/2381 on the gender balance among directors of listed companies (women on boards).* Official Journal of the European Union. (https://eur-lex.europa.eu/eli/dir/2022/2381)

European Parliament and Council (2022b). *Directive (eu) 2022/2464 (corporate sustainability reporting directive).* Official Journal of the European Union.

European Parliament and Council (2023). *Directive (eu) 2023/970 (pay transparency directive).* Official Journal of the European Union.

European Union (2020). *Regulation (eu) 2020/852 (eu taxonomy regulation).* Official Journal of the European Union.

Kanyamukenge, N., et al. (2024). [titolo su organisational testimonials]. *[Rivista]*, ,

McLaughlin, N., et al. (2021). [titolo su formati semantici machine-readable / x2rl]. *[Rivista]*, ,

Motzfeldt, N., et al. (2018). [titolo su digital-ready legislation]. *[Rivista]*, ,

Northcutt, C.G., et al. (2017). [titolo su confident learning / dataset cleaning]. *[Rivista]*, ,

Paruchuri, V. (2024). *Marker: Convert PDF to markdown quickly with high accuracy.* https://github.com/VikParuchuri/marker. (Software repository)

Rajpurkar, P., Zhang, J., Lopyrev, K., Liang, P. (2016). SQuAD: 100,000+ questions for machine comprehension of text. *Proceedings of the 2016 conference on empirical methods in natural language processing (emnlp)* (pp. 2383–2392).

Roemmele, M., Bejan, C.A., Gordon, A.S. (2011). Choice of plausible alternatives: An evaluation of commonsense causal reasoning. *Aaai spring symposium on logical formalizations of commonsense reasoning.*

Sargsyan, N., & Damiani, N. (2025). [titolo dell'articolo sulle legends e il compliance paradox]. *[Rivista]*, ,

Storey, N., et al. (2025). [titolo su generative ai per la sostenibilità]. *[Rivista]*, ,

The Economist (2024). *[titolo articolo che riporta la previsione gartner sul middle management].* [URL]. ([Data di accesso])

Wang, A., Pruksachatkun, Y., Nangia, N., Singh, A., Michael, J., Hill, F., ... Bowman, S.R. (2019). SuperGLUE: A stickier benchmark for general-purpose language understanding systems.

Advances in neural information processing systems (neurips) (Vol. 32).

Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., Bowman, S.R. (2018). GLUE: A multi-task benchmark and analysis platform for natural language understanding. *Proceedings of the 2018 emnlp workshop blackboxnlp* (pp. 353–355).

Wattenberg, N., et al. (2025). SustainableQA: [sottotitolo]. *[Rivista o atti]*, ,

Zhao, J., Wang, T., Yatskar, M., Ordonez, V., Chang, K.-W. (2018). Gender bias in coreference resolution: Evaluation and debiasing methods. *Proceedings of the 2018 conference of the north american chapter of the association for computational linguistics (naacl)* (pp. 15–20).

A Prompts

Listing 2: Passage classification prompt.

```
# System Prompt
You are an expert policy analyst specializing in the EU Gender Equality
Strategy 2020-2025. Your task is to accurately classify text chunks into
predefined thematic pillars. You must output only the exact category
label. Do not include any explanations, reasoning, punctuation, or formatting.

# User Prompt Template:

Content: {content}

Based primarily on the Content provided, classify this text chunk into one of
the following categories. Return only the single most appropriate class label
from the list below.

Categories:
1) Freedom from Violence and Stereotypes: Focuses on eliminating
gender-based violence, preventing physical/sexual violence, addressing online
harassment, ratifying the Istanbul Convention, and tackling gender bias in AI
and media. Label: violence_stereotypes

2) Thriving in a Gender-Equal Economy: Focuses on economic independence,
closing labor/pay/pension gaps, Work-Life Balance Directive, pay transparency,
women in STEM/entrepreneurship, and childcare (Barcelona targets). Label:
equal_economy

3) Leading Equally Throughout Society: Focuses on women's representation
in leadership (politics, agencies, boards), inclusive leadership, Women on
Boards Directive, and gender balance in management/elections. Label:
leadership_participation

4) Gender Mainstreaming and Intersectional Perspective: Focuses on
integrating gender into all EU policies (green transition, digital, health),
the Task Force for Equality, intersectionality (disability, migrant status,
age), and specific initiatives like the Green Deal or Cancer Plan. Label:
mainstreaming_intersectionality

5) Funding and Global Action for Equality: Focuses on financial mechanisms
(MFF, European Social Fund Plus, Horizon Europe) and global empowerment (GAP
III, international partnerships, trade). Label: funding_global_action

6) Unknown: Encompasses text that does not directly relate to the specific
pillars above. Includes semantically irrelevant text, administrative metadata,
headers, or general information outside the defined scopes. Label: unknown
```

Listing 3: Legend-generation system prompt.

```
You are a legend generator for an academic experiment on EU regulatory compliance.

A ``legend" is a short fictional story featuring named characters who demonstrate
concrete compliance with a specific aspect of the EU Gender Equality Strategy
2020-2025. Every legend you produce must follow this exact structure:

# Title -- A short, evocative title.
```

```
## Setting -- Organisation, sector, country. Vary across legends.
## Characters -- 3-5 named characters with roles and a one-line description.
## The Story -- 400-600 word narrative with at least one direct-speech
                    dialogue, a concrete gender-equality gap addressed by
                    specific measures, ending with a measurable positive
                    outcome and an acknowledgement of the Strategy.

Content rules:
 - Focus exclusively on the regulation point provided by the user.
 - Cite "EU Gender Equality Strategy 2020-2025" at least once.
 - Professional but accessible tone.
 - Do not reuse sectors, countries, or character names across legends.
 - Make compliance actions concrete and realistic (salary audits, mentorship
   programmes, policy changes, training workshops) -- never generic or vague.
```

Listing 4: Span Extractor prompt (Stage 1).

```
You are a high-recall span extractor specialised in the EU Gender Equality
Strategy 2020-2025. Identify every verbatim segment of text that aligns
with the strategy's pillars, legal directives, or socio-economic targets.

TASK: Extract concise, verbatim spans relevant to the Classification below.
INPUT: Context (passage), Classification (pillar label).

EXTRACTION SCOPE (PILLARS):
 1. Gender-Based Violence
 2. Gender Stereotypes
 3. Labour Market & Care Gap
 4. Equal Participation
 5. Gender Pay & Pension Gap
 6. Decision-Making Balance

EXTRACTION RULES:
 - Verbatim Extraction: do not paraphrase.
 - Legal & Technical Terms: prioritise directives (Pay Transparency Directive,
   Women on Boards) and quantitative targets (40% non-executive directors).
 - Span Length: 1-5 words typical; up to 8 for complex named entities.
 - Exclude generic organisational filler.

OUTPUT: { "spans": ["<span 1>", "<span 2>", ...] }
```

Listing 5: Span Filtering and Thematic Clustering prompt (Stage 2).

```

SYSTEM: You are a highly precise data structuring assistant. Your SOLE task is to
critically evaluate candidate spans, filter them based on context and rules,
group the valid ones, and output ONLY a valid JSON object in the specified
format. Adherence to all rules and the JSON format is critical.


TASK: You are an expert grouping and filtering assistant. Your goal is to organize
spans into meaningful, specific thematic groups, minimizing use of the
"Individual" category.

Instructions:
  Follow these steps sequentially:
     1. Filter & Consolidate:
         - Keep only spans found verbatim in Context (case-insensitive, original casing).
         - Remove trivial/generic standalone spans.
         - If spans are redundant (substrings, acronym/expansion, exact synonyms conveying
             the same core idea), keep only the single most complete and informative version
             . Discard others.

     2. Thematic Grouping (Primary Focus):
         - Actively group spans sharing a clear, specific, and meaningful common theme or
             semantic relationship.
         - Form groups even with 2-3 highly related spans. Do not default to "Individual" if
              a small, specific group can be formed.
         - Create a concise (2-5 words), highly specific `label` for each group capturing
             its core theme. Avoid generic labels.
         - Prioritize creating thematic groups.

     3. "Individual" Category (Use Sparingly):
         - Only spans that are truly disparate and cannot be thematically linked to any
             other span (even in a small group) go into the group labeled `"Individual"`.
```

```
        - Before assigning to "Individual", double-check if it could form a small thematic
            group with another span.
    4. JSON Output: Return ONLY one valid JSON object in this exact format. No extra text.
        {{
          "groups": [
            {{
              "label": "<Group Label>",
              "spans": ["<Span 1>", "<Span 2>"]
            }},
            {{
              "label": "Individual",
              "spans": ["<Span>"]
            }}
          ]
          }}
Context: "{content}"
Classification: "{classification}"
Candidate Spans:"{candidate_lines}"
```

Listing 6: Factoid Q&A generation prompt.

```
System Prompt
You are an expert Q\&A generator. You create factoid questions strictly based on provided
    text and spans, ensuring the spans are the exact verbatim answers. You output only valid
     JSON.


User Prompt
TASK: Generate high-quality, factoid Question-Answer pairs based strictly and solely on the
    provided Context.

INPUT DATA:
- Context: "{content}" - The source text.
- Spans (JSON): {json.dumps(spans)} - Specific text excerpts from Context; these are the
    ONLY valid answers.
- Classification: {classification} - Required tag for output pairs.

CORE RULES (APPLY TO ALL GENERATION - NON-NEGOTIABLE):
1. Answer Validity: The answer field MUST be EXACT, VERBATIM SPAN(S) ONLY from the input
    Spans (JSON). NO additions, omissions, paraphrasing, or changes allowed.
2. Contextual Accuracy: The answer MUST be the PRECISE, COMPLETE, CORRECT response to the
    question, based STRICTLY AND SOLELY on the provided Context.
3. Conditional Generation: Generate a Q\&A pair ONLY IF you can form a natural, clear
    question for which the span(s) are the perfect, complete, and verbatim answer as found
    in the Context, AND all Core Rules are met. If any rule is violated, DO NOT generate the
     pair. Prioritize accuracy over quantity.
4. Metadata: Each output object MUST include "type": "Factoid" and "tag": "{classification
    }".

GENERATION PROCESS:
- Think step-by-step internally to identify valid Q\&A opportunities based on the Spans and
    Context. Do not output intermediate thoughts.
- Span Handling Strategy:
    - For each group in Spans (JSON) (excluding "Individual"):
        - create one question answered by ALL spans in the group combined (e.g., "SpanA,
            SpanB"). The combined answer format should typically use comma+space if it fits
            the context. Must meet Core Rules.
        - create one separate question for EACH INDIVIDUAL span within the group. Must meet
            Core Rules.
    - For each individual span in Spans (JSON) ("Individual" category):
        - create one question for that single span. Must meet Core Rules.
- Question Formulation:
    - Questions must be grammatical, natural, standalone, and vary in structure (What, When,
        Which, Percentage, Value, etc.).
    - Reference organization name from Context if applicable.

EXPLICITLY AVOID (DO NOT GENERATE THESE QUESTION TYPES):
- Trivial questions (answer obvious from span alone).
- Questions embedding the answer text.
- "What is [span]?" questions unless Context gives an explicit definition.
- "Why" or "How" questions.

MANDATORY FINAL VALIDATION:
- Before finalizing output, review EVERY potential pair generated against the Core Rules and
     Explicitly Avoid list:
    1. Does answer meet Rule 1 (EXACT VERBATIM SPAN(S) ONLY)? (Y/N)
```

```
   2. Does answer meet Rule 2 (PRECISE, COMPLETE, CORRECT per CONTEXT ONLY)? (Y/N)
   3. Does question meet Rule 3 (natural/clear) and avoid all types listed under "
       Explicitly Avoid"? (Y/N)
- DISCARD any pair failing any check. Only include pairs where ALL checks pass.

OUTPUT FORMAT (JSON ONLY):
Output ONLY a valid JSON object {{"qa_pairs": [...]}}. If none qualify, output {{"qa_pairs":
     []}}.
Each object in qa_pairs MUST have: "question" (string), "answer" (string), "type": "Factoid
    ", "tag": "{classification}".


Example JSON Structure:
   {{
     "qa_pairs": [
       {{
         "question": "<Your Well-Formed Question>",
         "answer": "<Exact Span(s) Required>",
         "type": "Factoid",
         "tag": "{classification}"
       }}
     ]
   }}
```

Listing 7: Non-Factoid Q&A generation prompt.

```
System Prompt:
You are an expert Q\&A generator. You create descriptive, non-factoid questions and concise
    answers based strictly on the provided text. You output only valid JSON and meticulously
     follow all formatting and content rules.


User Prompt:
Task: Generate high-quality, distinct, non-factoid (descriptive/explanatory) Question-Answer
     pairs STRICTLY from the provided Context. Aim for 7-15 pairs, or more if the Context
    supports them without sacrificing quality or distinctness.

Inputs:
- Context: "{content}"
- Classification: "{classification}"

Key Instructions \& Constraints:
1. Strict Context Adherence: ALL questions and answers MUST be derived SOLELY from
    information EXPLICITLY STATED or DIRECTLY INFERABLE within the Context. NO external
    knowledge or assumptions.
2. Non-Factoid Focus:
   - Questions MUST require descriptive or explanatory answers (e.g., understanding
       relationships, purposes, definitions, processes, implications as presented in the
       text).
   - AVOID simple factoid questions answerable by a single data point (date, number, short
       proper noun unless it's being defined/explained).
3. Distinctness: Each Q\&A pair must offer a unique insight or cover a different descriptive
     aspect, relationship, process, or explanation from the Context. Avoid redundancy.
4. Answerability: Every question MUST be fully answerable using ONLY the Context.

Question Formulation Guidance:
- Frame questions to elicit explanations, descriptions, or understanding of:
   - Relationships (e.g., 'How does X relate to Y, according to the text?')
   - Processes/Mechanisms (e.g., 'Explain the mechanism for X based on the context.')
   - Reasons/Justifications (e.g., 'What reasons are provided for Z in the text?')
   - Implications/Consequences (e.g., 'What are the implications of X as detailed?')
   - Definitions/Clarifications of concepts within the text.
- For 'Why'/'How': Ask for explanations of purpose, reasoning, methods, or processes as
    understood from the text (e.g., 'Explain the rationale behind X...' or 'Describe the
    method for X...').
- If a complex topic has multiple facets (e.g., several reasons, steps in a process),
    generate distinct Q\&A for substantial components.
- Use company names from Context for specificity if available.

Answer Formulation Guidance:
- Answers must be comprehensive yet focused, accurately summarizing/explaining relevant
    Context information.
- Aim for 1-4 sentences and approximately 25 to 70 words long, but ensure the answer fully
    addresses the question based ONLY on the text, even if slightly longer. Avoid trivial/
    short-phrase answers.
```

```
Mandatory Self-Correction Checklist (Review EACH pair before output):
1. Is the question genuinely non-factoid (descriptive/explanatory)? (Y/N)
2. Is the Q\&A pair distinct from others? (Y/N)
3. Is the question fully answerable ONLY from Context? (Y/N)
4. Is the answer comprehensive, accurate, and derived ONLY from Context? (Y/N)
- DISCARD any pair failing ANY check.

Output Format (JSON ONLY):
Return ONLY one valid JSON object: {{"qa_pairs": [...]}}. If no pairs qualify, use {{"
    qa_pairs": []}}.
Each object in qa_pairs MUST have:
- "question": (string)
- "answer": (string)
- "type": "Descriptive, non-factoid"
- "tag": "{classification}"

Example:
{{
  "qa_pairs": [
    {{"question": "...", "answer": "...", "type": "Descriptive, non-factoid", "tag": "{
        classification}"}}
  ]
}}
```

B Examples

Q&A Examples

Factoid:

```
{
  "question": "What type of enterprises were established by women who
  participated in the green economy empowerment initiative?",
  "answer": "two women-led solar-maintenance micro-enterprises",
  "type": "Factoid",
  "tag": "equal_economy"
}
```

Non-Factoid:

```
{
  "question": "Explain how Hugo addressed the issue of disincentives for second
  earners.",
  "answer": "Hugo launched an internal campaign explaining how
  national tax and benefit systems can create disincentives for second earners.
  The campaign was meant to help employees understand their rights and options
  under the new framework.",
  "type": "Descriptive, non-factoid",
  "tag": "equal_economy"
}
```

JSONL Training Line

```
{
  "messages": [
    {
      "role": "system",
      "content": "You are an expert assistant on the EU Gender Equality Strategy 2020-2025. Answer questions
          ..."
    },
    {
      "role":
      "user", "content": "How does the passage describe the foundation on which a prosperous and social
          Europe rests?"
    },
    {
      "role": "assistant",
      "content": "The passage states that a prosperous and social Europe depends
      on everyone, framing economic and social progress as inseparable from
      collective participation. It then outlines the equal opportunities and
      economic independence that women and men in all their diversity should
      enjoy."
    }
  ]
}
```

GE-NLI Item


```
{
  "id": "ge_nli_001",
  "premise": "FinBelge S.A., a Belgian investment firm, ...",
  "hypothesis": "Listed companies must ensure that at least 40% ...",
  "label": "entailment",
  "pillar_premise": "leadership_participation",
  "pillar_hypothesis": "leadership_participation",
  "source_passage_id": "WomenOnBoards_Dir_2022_2381__c004",
  "construction_method": "llm_generated_premise"
}
```


GE-NEXT Item (Inés Lobato vignette)

Vignette. Inés Lobato, Head of People at a Madrid logistics company with 380 employees, has completed the company's first pay audit following the transposition of the Pay Transparency Directive. The audit reveals an unexplained 9% gender pay gap concentrated in the operations division, affecting 42 women. Inés must propose a remediation plan to the board.

A. Issue a company-wide statement reaffirming the company's commitment to fair pay and equal opportunities. *(performative)*
B. Defer remediation to the following financial year to spread the cost across two budget cycles, reporting the gap in the annual sustainability report as "under review". *(cost-optimising)*
C. Build a banded 24-month remediation plan with quarterly salary adjustments, individual notification of affected employees and quarterly progress disclosure to the works council, in line with the reporting obligations of the Pay Transparency Directive. *(substantive-compliant,* ***gold)***
D. Commission a six-month external diversity training programme for the management team of the operations division. *(orthogonal)*

Label: `C`